How to design a CFG for a given language?

- Designing context-free grammar is tricky.
- You need to be able to identify recursive structure in your language
- Some CFL are union of simpler CFLs.
- we can start by constructing CFG for smaller CFL and then merge them together.

Design a context free grammar.

$\Sigma = \{a, b\}$

Ex 01: $A = \{ w \mid \text{the length of } w \text{ is even} \}$

ε, a, b, aa, ab, ba, bb, ...
✓ ✗ ✗ ✓ ✓ ✓ ✓

S

$S \rightarrow aSa \mid aSb \mid bSa \mid bSb \mid \varepsilon$

$S \rightarrow aaS \mid abS \mid baS \mid bbS \mid \varepsilon$

$S \rightarrow aaS \mid abS \mid baS \mid bbS \mid T$

$T \rightarrow \varepsilon$

Ex 02: $B = \{ w \mid w$ starts and ends with the same symbol $\}$

$\Sigma = \{a, b\}$

ε, a, b, a^*, b^*, a __ b, b __ a

✓ ✓

a __ a ✓

b __ b ✓

$S \longrightarrow aBa \mid bBb \mid a \mid b \mid \varepsilon$

$B \longrightarrow aB \mid bB \mid \varepsilon$

Ex3! $C = \{ w \mid w \text{ contains twice as many a's as b's} \}$

$\Sigma = \{a, b\}$

ε ✓, ab ✗, aab ✓, aba ✓, baa ✓, a ✗

$$S \to SaSaSbS \mid SaSbSaS \mid SbSaSaS \mid \varepsilon$$

$B = \{ w \mid w \text{ has at least 3 a's} \}$

◯ a ◯ a ◯ a ◯

$$S \to TaTaTaT$$

$$T \to aT \mid bT \mid \varepsilon$$

* Remember
  - All regular languages can be expressed by context free grammar
  - Essentially, CFG is more powerful than regular expressions.

Ambiguity

Sometimes a grammar can generate the same string in several different ways.

Such strings will have several different parse trees.

In some applications, this may be undesirable.

If a grammar generates a string in several different ways, we say that string is derived ambiguously in that grammar.

If a grammar generates some string ambiguously, we say that the grammar is ambiguous.

Example:

$$\langle E \rangle \rightarrow \langle E \rangle + \langle E \rangle \mid \langle E \rangle \times \langle E \rangle \mid (\langle E \rangle) \mid a$$

$$V = \{ \langle E \rangle \} \qquad \Sigma = \{ a, (,), +, \times \}$$

$$S = \langle E \rangle$$

$$a + a \times a$$

1. $\langle E \rangle \Rightarrow \langle E \rangle + \langle E \rangle \Rightarrow \langle E \rangle + \langle E \rangle \times \langle E \rangle$

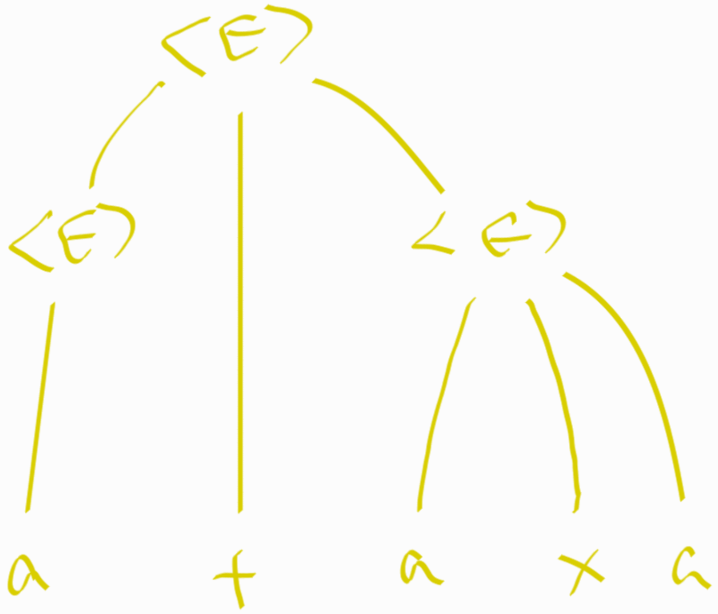
$$\Downarrow^{*}$$

$$a + a \times a$$

2. $\langle E \rangle \Rightarrow \langle E \rangle \times \langle E \rangle \Rightarrow \langle E \rangle + \langle E \rangle \times \langle E \rangle$

$$\Downarrow^{*}$$

$$a + a \times a$$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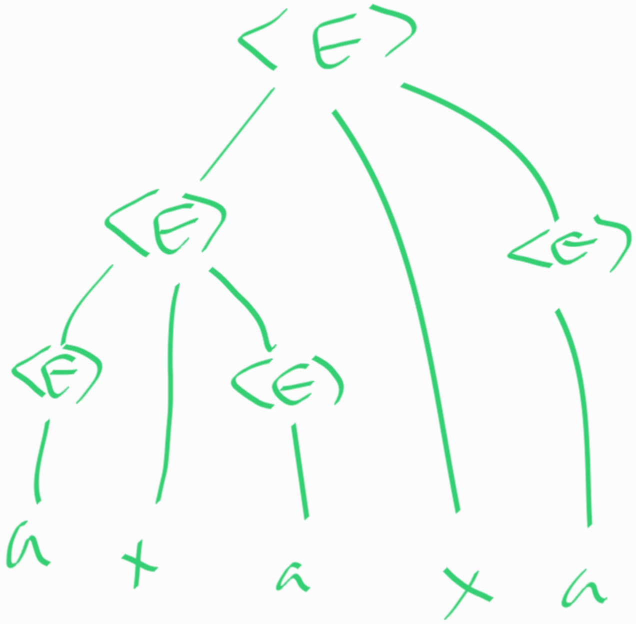
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